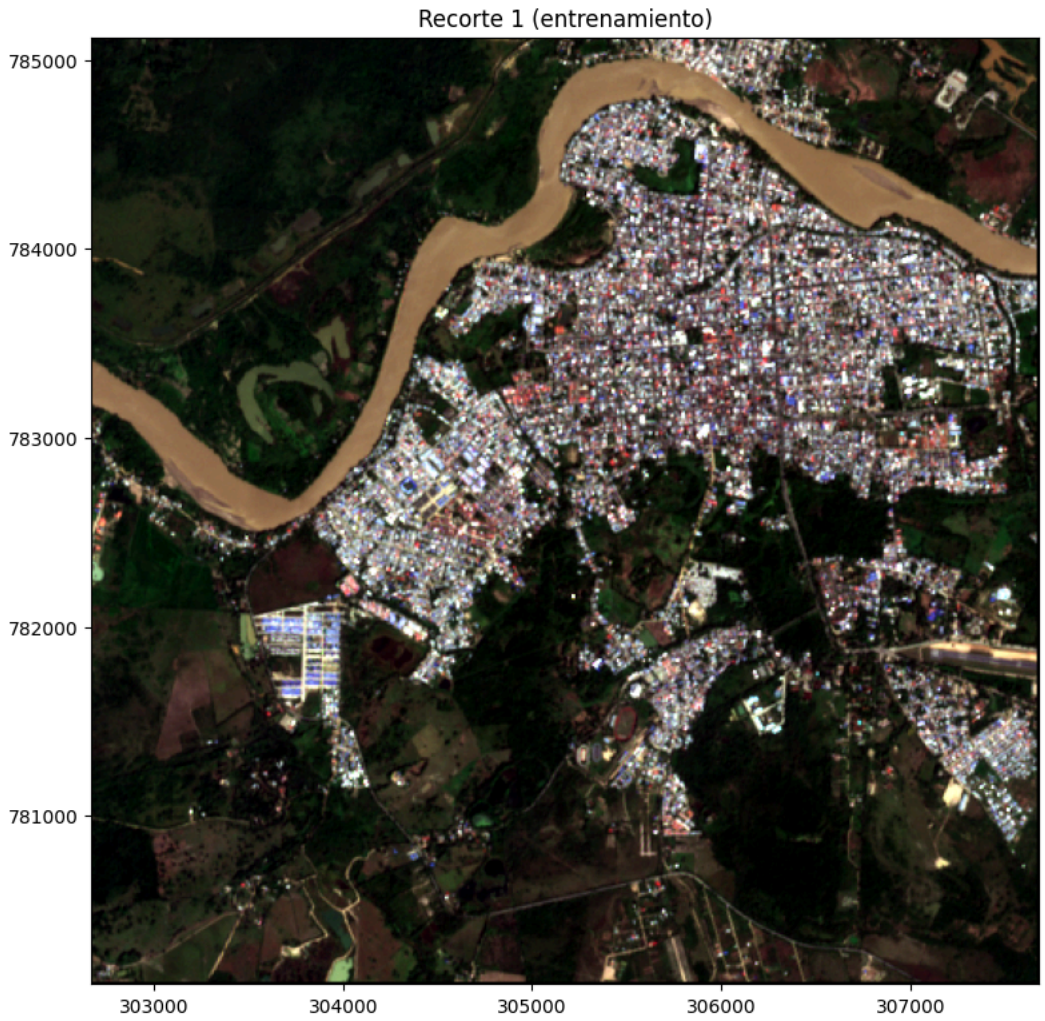
```
!pip install -q rasterio geopandas scikit-learn scipy matplotlib seaborn gdown
```

```
!wget -q -O entregaclasificacion.py https://raw.githubusercontent.com/AlexandraUD/ClasificacionSupervisada/main/entregaclasific
```

```
%run entregaclasificacion.py
```

```
======================================================================
PASO 1: Recortes de la imagen Sentinel-2
======================================================================
Ya existe s2_recorte1.tif, no se vuelve a generar.
Ya existe s2_recorte2.tif, no se vuelve a generar.
```



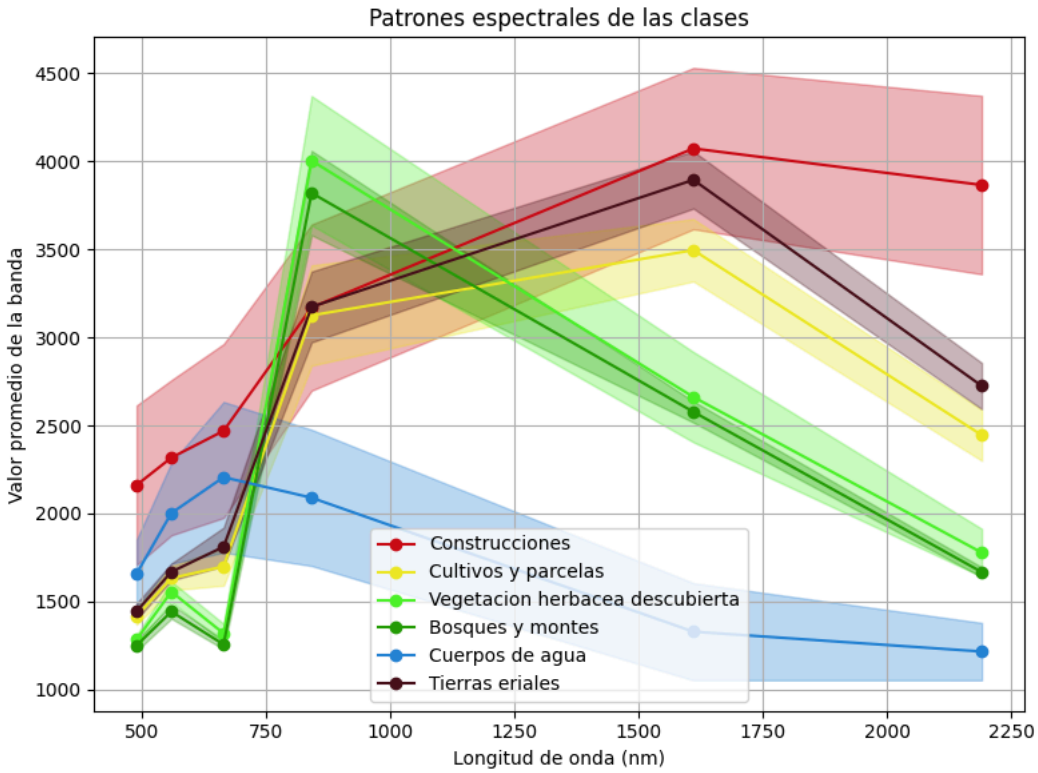


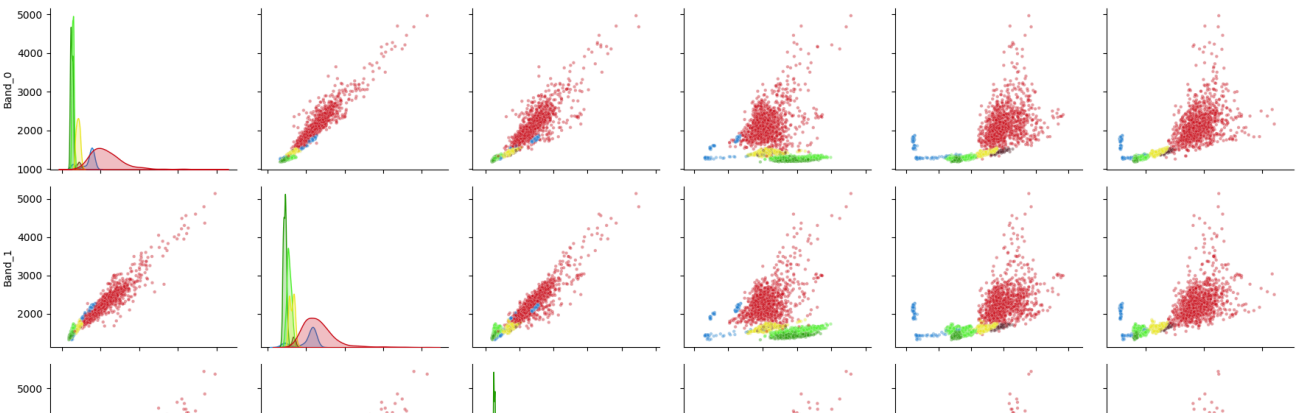


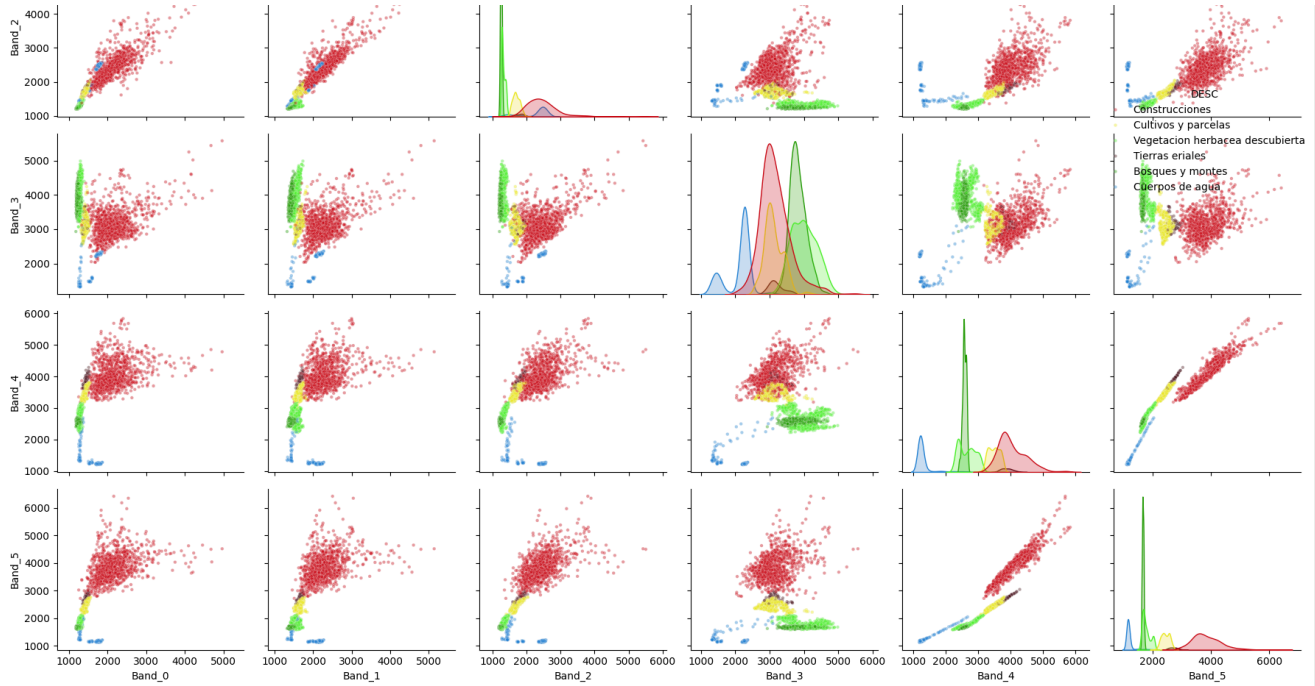

```
======================================================================
PASO 2: Muestras de entrenamiento (recorte 1)
======================================================================
Clases: {1: 'Construcciones', 2: 'Cultivos y parcelas', 3: 'Vegetacion herbacea descubierta', 4: 'Bosques y montes', 5: 'Cuerpos

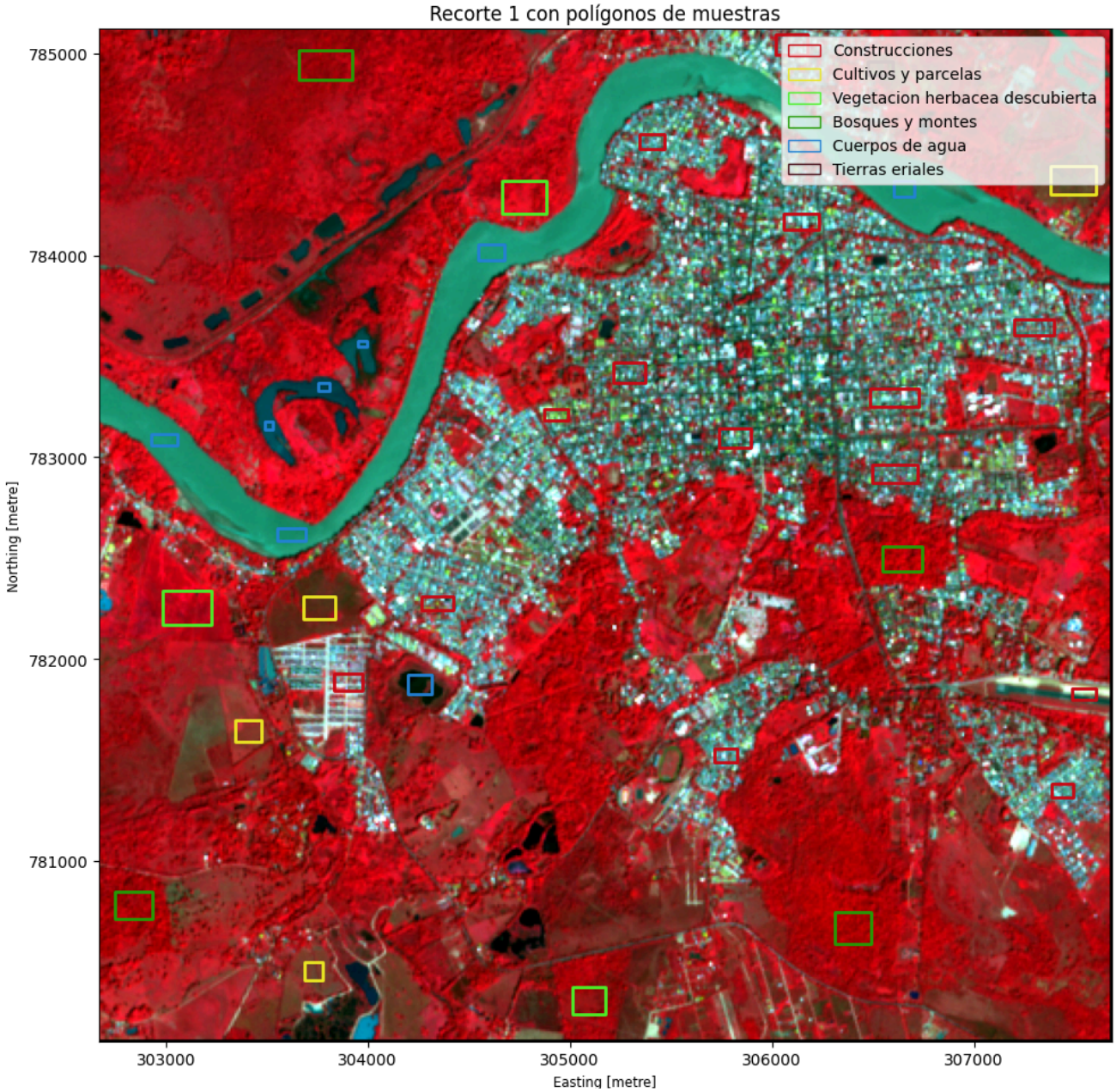
Muestras recorte 1 por clase:
Clase  DESC
1      Construcciones                     1814
2      Cultivos y parcelas                 708
3      Vegetacion herbacea descubierta    1015
4      Bosques y montes                   1172
5      Cuerpos de agua                     495
6      Tierras eriales                      84

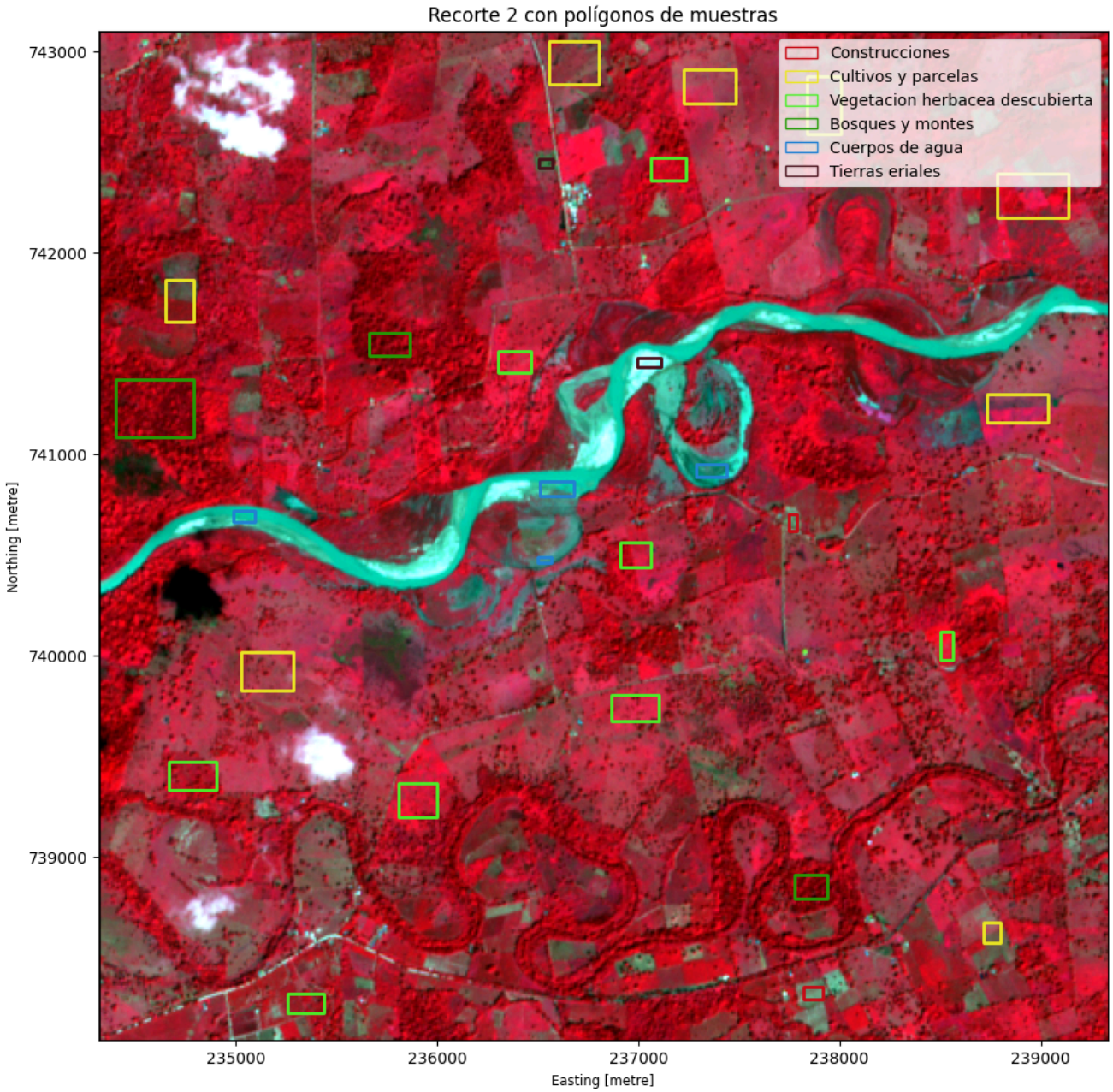
Muestras recorte 2 por clase:
Clase  DESC
1      Construcciones                       92
2      Cultivos y parcelas                3572
3      Vegetacion herbacea descubierta    1744
4      Bosques y montes                   1527
5      Cuerpos de agua                     313
6      Tierras eriales                      90
```



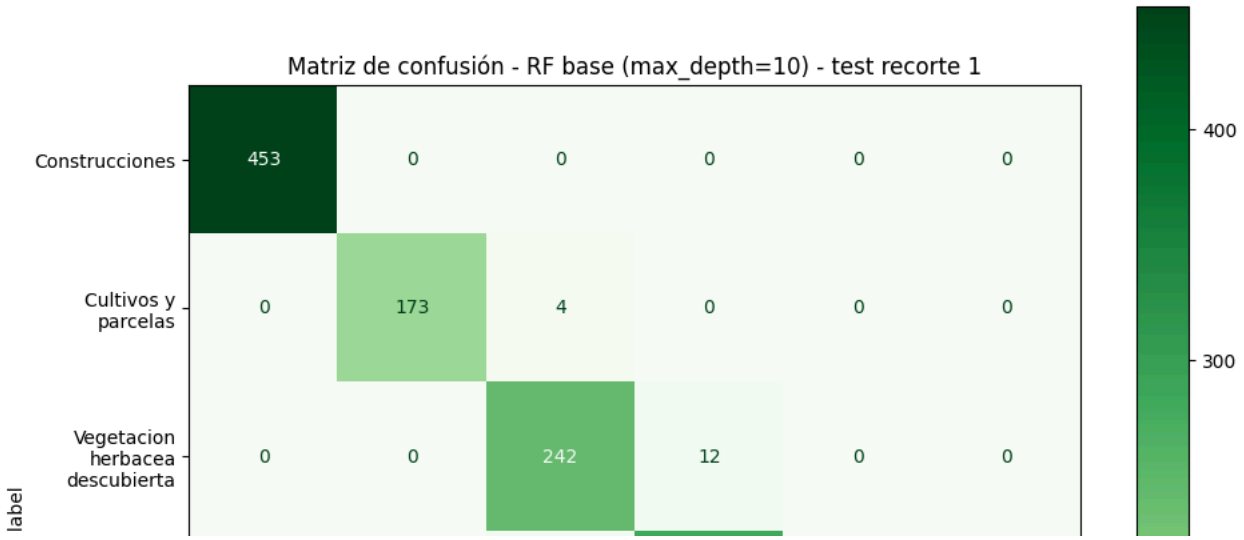


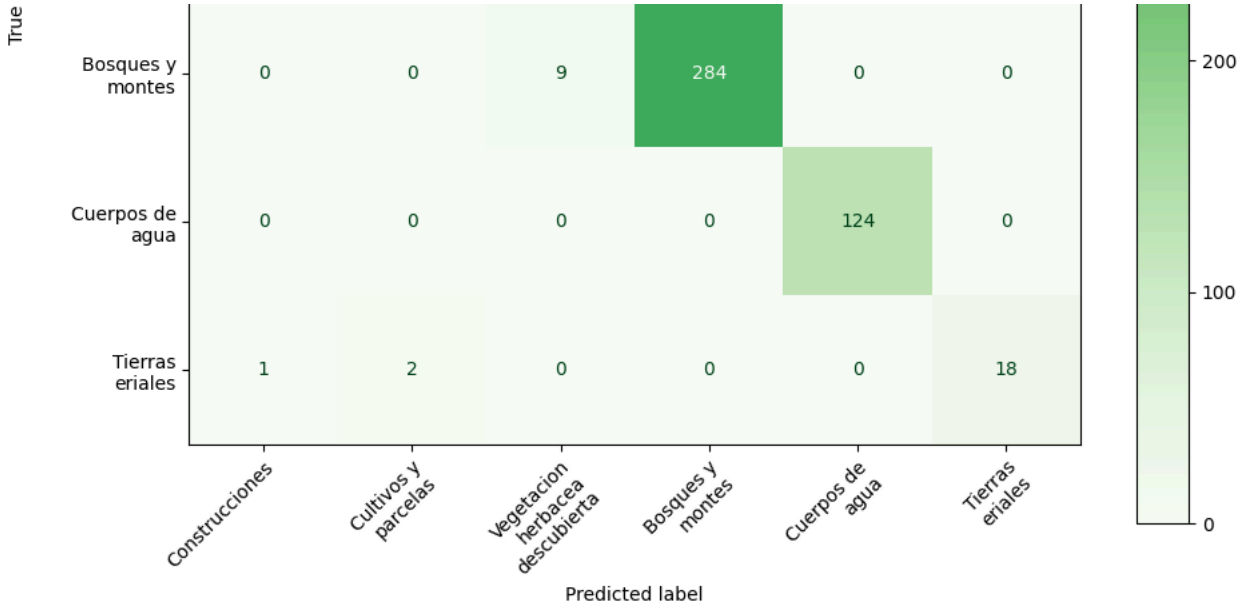

Band_2
Band_3
Band_4
Band_5
Band_0
Band_1
Band_2
Band_3
Band_4
Band_5
DESC
Construcciones
Cultivos y parcelas
Vegetacion herbacea descubierta
Tierras eriales
Bosques y montes
Cuerpos de agua


Recorte 1 con polígonos de muestras
Construcciones
Cultivos y parcelas
Vegetacion herbacea descubierta
Bosques y montes
Cuerpos de agua
Tierras eriales
Northing [metre]
785000
784000
783000
782000
781000
303000
304000
305000
306000
307000
Easting [metre]




```
======================================================================
PASO 3: Random Forest
======================================================================
Tamaño de y_train: (3966,) | Tamaño de y_test: (1322,)
```






```
----- RF base (max_depth=10) - test recorte 1 -----
                              precision    recall  f1-score   support

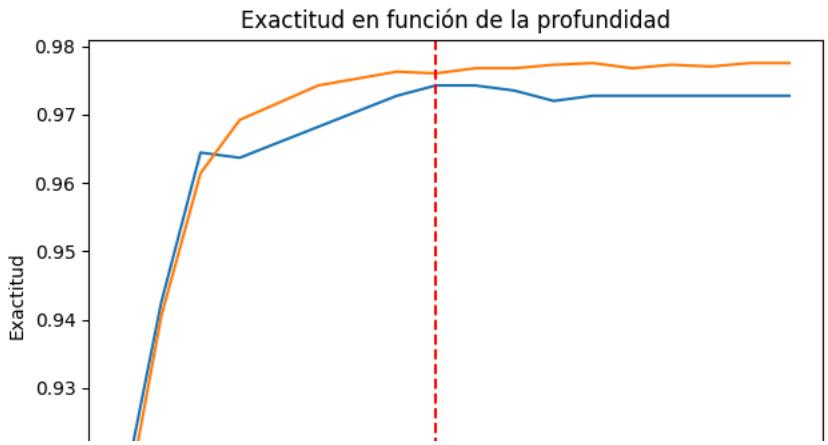
                Construcciones       1.00      1.00      1.00       453
            Cultivos y parcelas       0.99      0.98      0.98       177
Vegetacion herbacea descubierta       0.95      0.95      0.95       254
              Bosques y montes       0.96      0.97      0.96       293
               Cuerpos de agua       1.00      1.00      1.00       124
               Tierras eriales       1.00      0.86      0.92        21

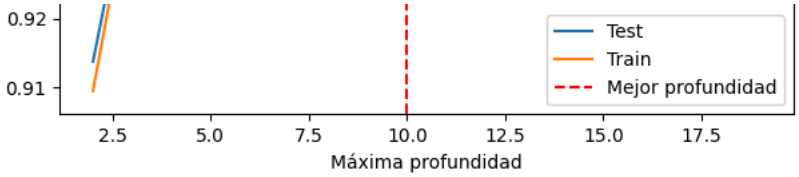
                      accuracy                           0.98      1322
                     macro avg       0.98      0.96      0.97      1322
                  weighted avg       0.98      0.98      0.98      1322

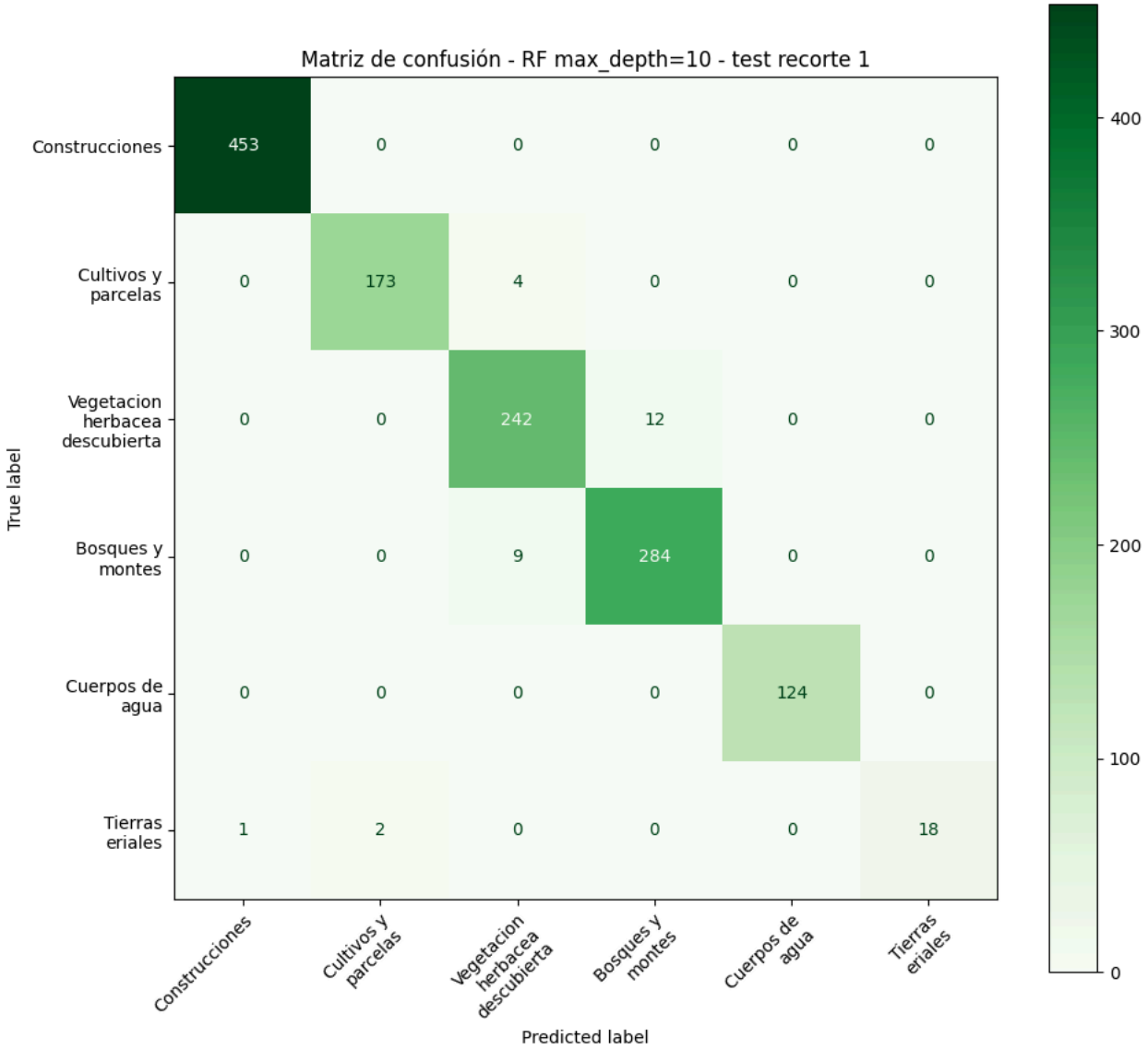
Precisión global: 0.979
Índice kappa: 0.972

Buscando el max_depth óptimo (2 a 19)...
  max_depth =  2 | CV test = 0.914 | CV train = 0.909
  max_depth =  3 | CV test = 0.943 | CV train = 0.940
  max_depth =  4 | CV test = 0.964 | CV train = 0.961
  max_depth =  5 | CV test = 0.964 | CV train = 0.969
  max_depth =  6 | CV test = 0.966 | CV train = 0.972
  max_depth =  7 | CV test = 0.968 | CV train = 0.974
  max_depth =  8 | CV test = 0.970 | CV train = 0.975
  max_depth =  9 | CV test = 0.973 | CV train = 0.976
  max_depth = 10 | CV test = 0.974 | CV train = 0.976
  max_depth = 11 | CV test = 0.974 | CV train = 0.977
  max_depth = 12 | CV test = 0.974 | CV train = 0.977
  max_depth = 13 | CV test = 0.972 | CV train = 0.977
  max_depth = 14 | CV test = 0.973 | CV train = 0.978
  max_depth = 15 | CV test = 0.973 | CV train = 0.977
  max_depth = 16 | CV test = 0.973 | CV train = 0.977
  max_depth = 17 | CV test = 0.973 | CV train = 0.977
  max_depth = 18 | CV test = 0.973 | CV train = 0.978
  max_depth = 19 | CV test = 0.973 | CV train = 0.978

Mejor resultado: profundidad 10, exactitud 0.974
```





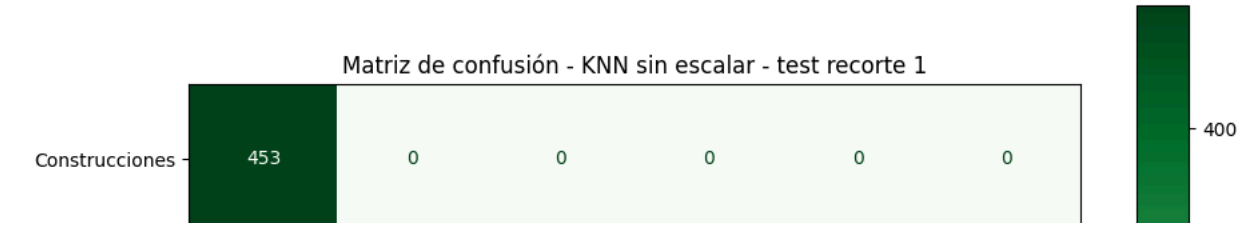


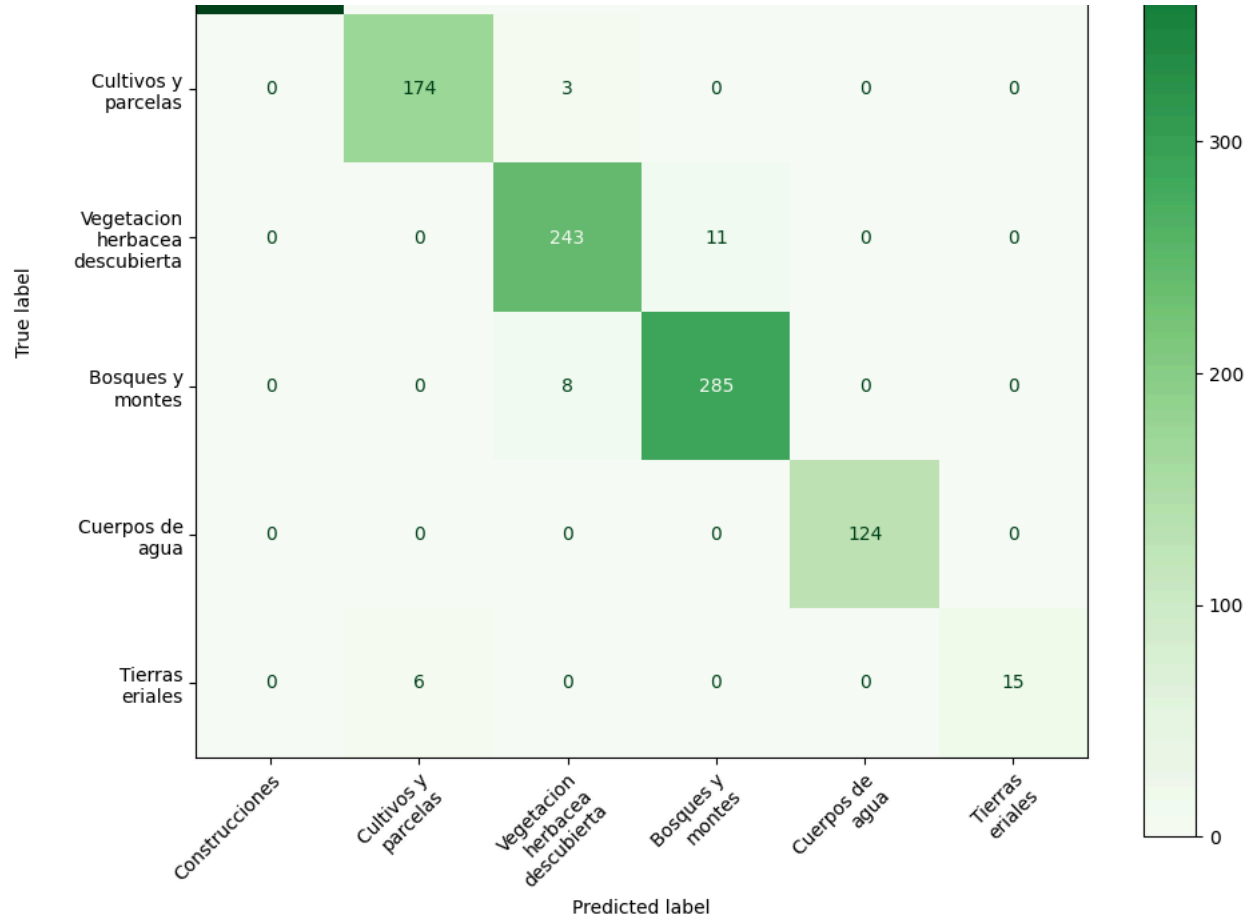

```
----- RF max_depth=10 - test recorte 1 -----
                            precision    recall  f1-score   support

                Construcciones       1.00      1.00      1.00       453
             Cultivos y parcelas       0.99      0.98      0.98       177
Vegetacion herbacea descubierta       0.95      0.95      0.95       254
               Bosques y montes       0.96      0.97      0.96       293
                Cuerpos de agua       1.00      1.00      1.00       124
                Tierras eriales       1.00      0.86      0.92        21

                       accuracy                           0.98      1322
                      macro avg       0.98      0.96      0.97      1322
                   weighted avg       0.98      0.98      0.98      1322

Precisión global: 0.979
Índice kappa: 0.972
```



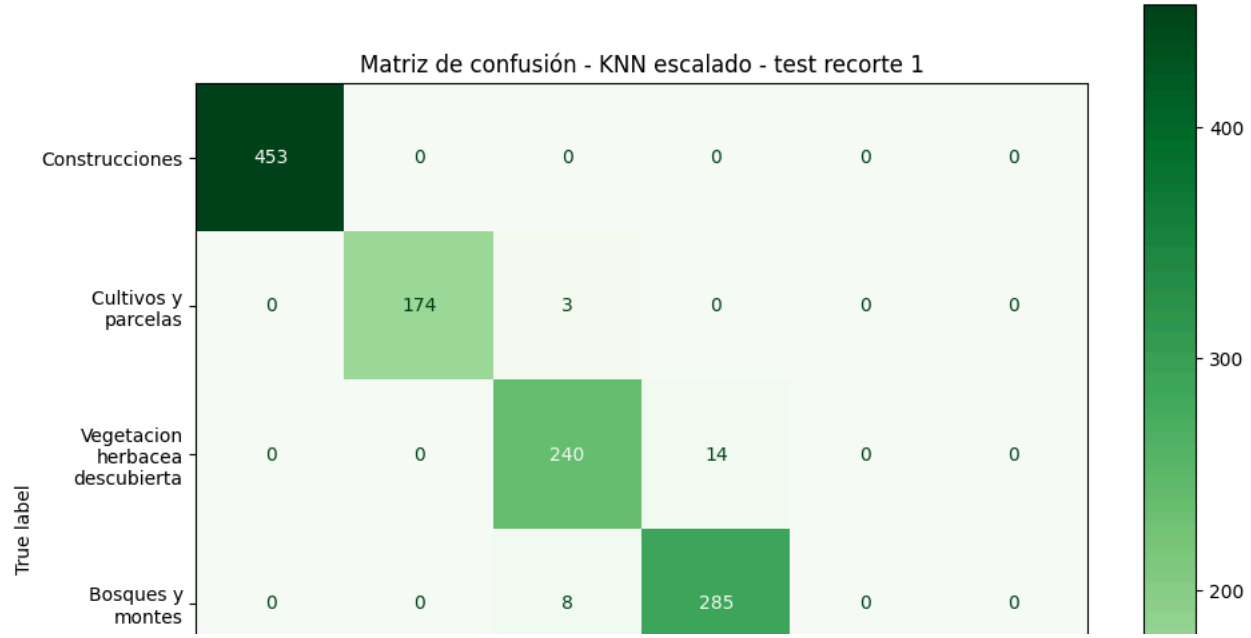


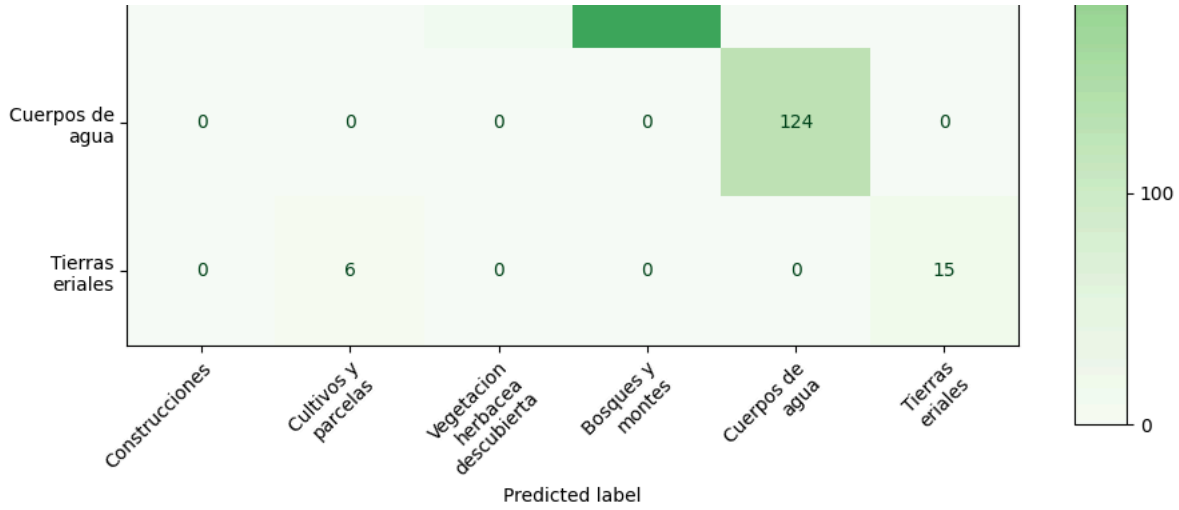

```
----- KNN sin escalar - test recorte 1 -----
                            precision    recall  f1-score   support

             Construcciones       1.00      1.00      1.00       453
          Cultivos y parcelas       0.97      0.98      0.97       177
Vegetacion herbacea descubierta       0.96      0.96      0.96       254
          Bosques y montes       0.96      0.97      0.97       293
           Cuerpos de agua       1.00      1.00      1.00       124
           Tierras eriales       1.00      0.71      0.83        21

                  accuracy                           0.98      1322
                 macro avg       0.98      0.94      0.96      1322
              weighted avg       0.98      0.98      0.98      1322

Precisión global: 0.979
Índice kappa: 0.972
```





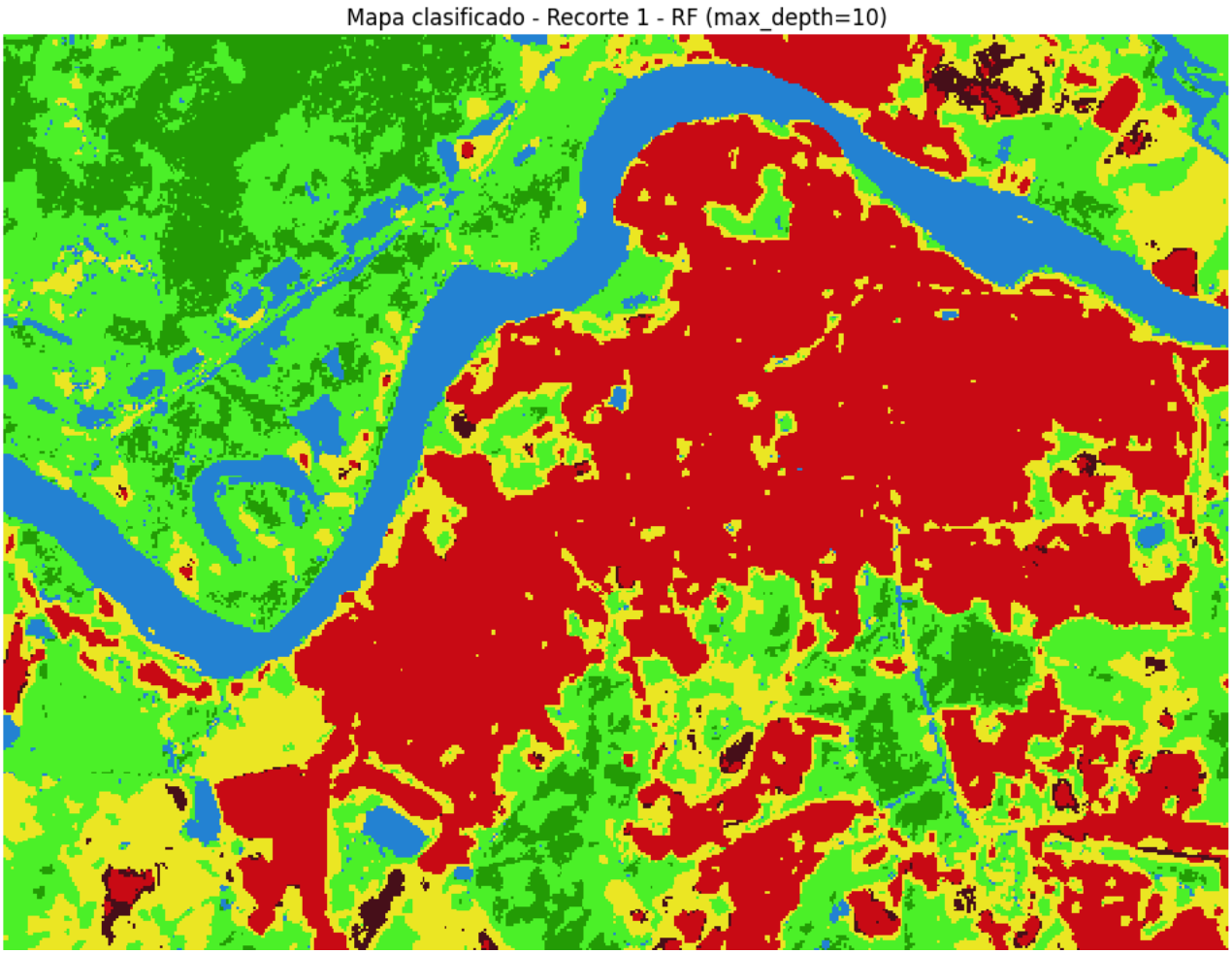

```
----- KNN escalado - test recorte 1 -----
                              precision    recall  f1-score   support

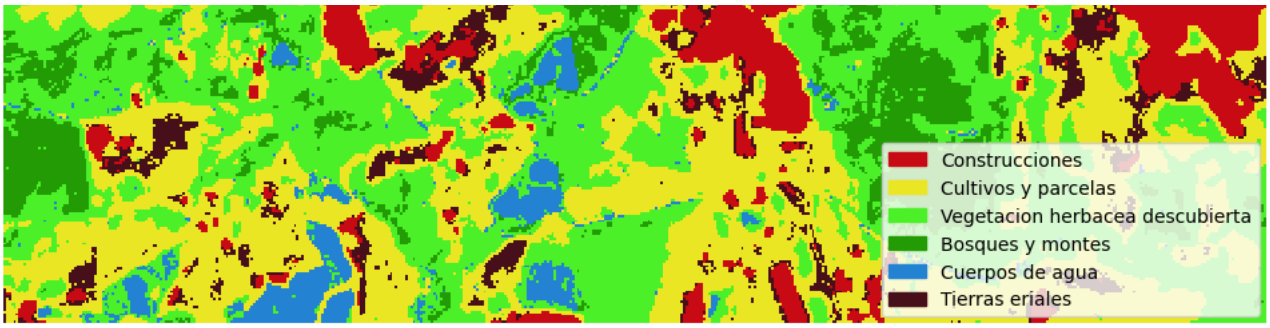
                Construcciones       1.00      1.00      1.00       453
            Cultivos y parcelas       0.97      0.98      0.97       177
Vegetacion herbacea descubierta       0.96      0.94      0.95       254
                Bosques y montes       0.95      0.97      0.96       293
                Cuerpos de agua       1.00      1.00      1.00       124
                Tierras eriales       1.00      0.71      0.83        21

                       accuracy                           0.98      1322
                      macro avg       0.98      0.94      0.95      1322
                   weighted avg       0.98      0.98      0.98      1322

Precisión global: 0.977
Índice kappa: 0.969

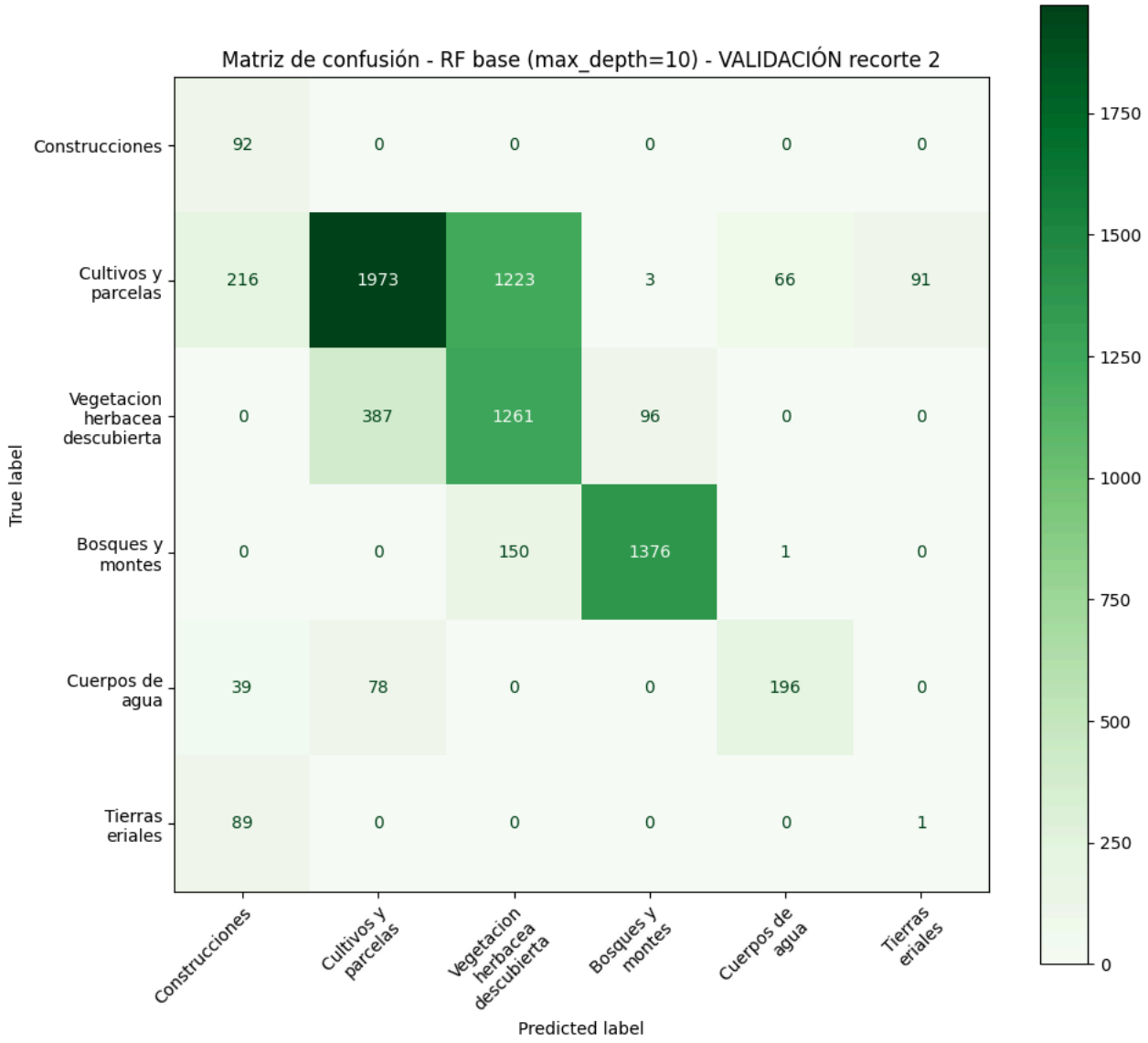
Clasificando el recorte 1...
  Valores únicos del mapa: [    1     2     3     4     5     6 65535]
```






```
Guardado: /content/drive/MyDrive/ClasificacionSupervisada/salidas/clasificacion_recorte1_rf.tif
Guardado: /content/drive/MyDrive/ClasificacionSupervisada/salidas/clasificacion_recorte1_rf.gpkg

======================================================================
PASO 4: Validación con el recorte 2 y muestras independientes
======================================================================
```

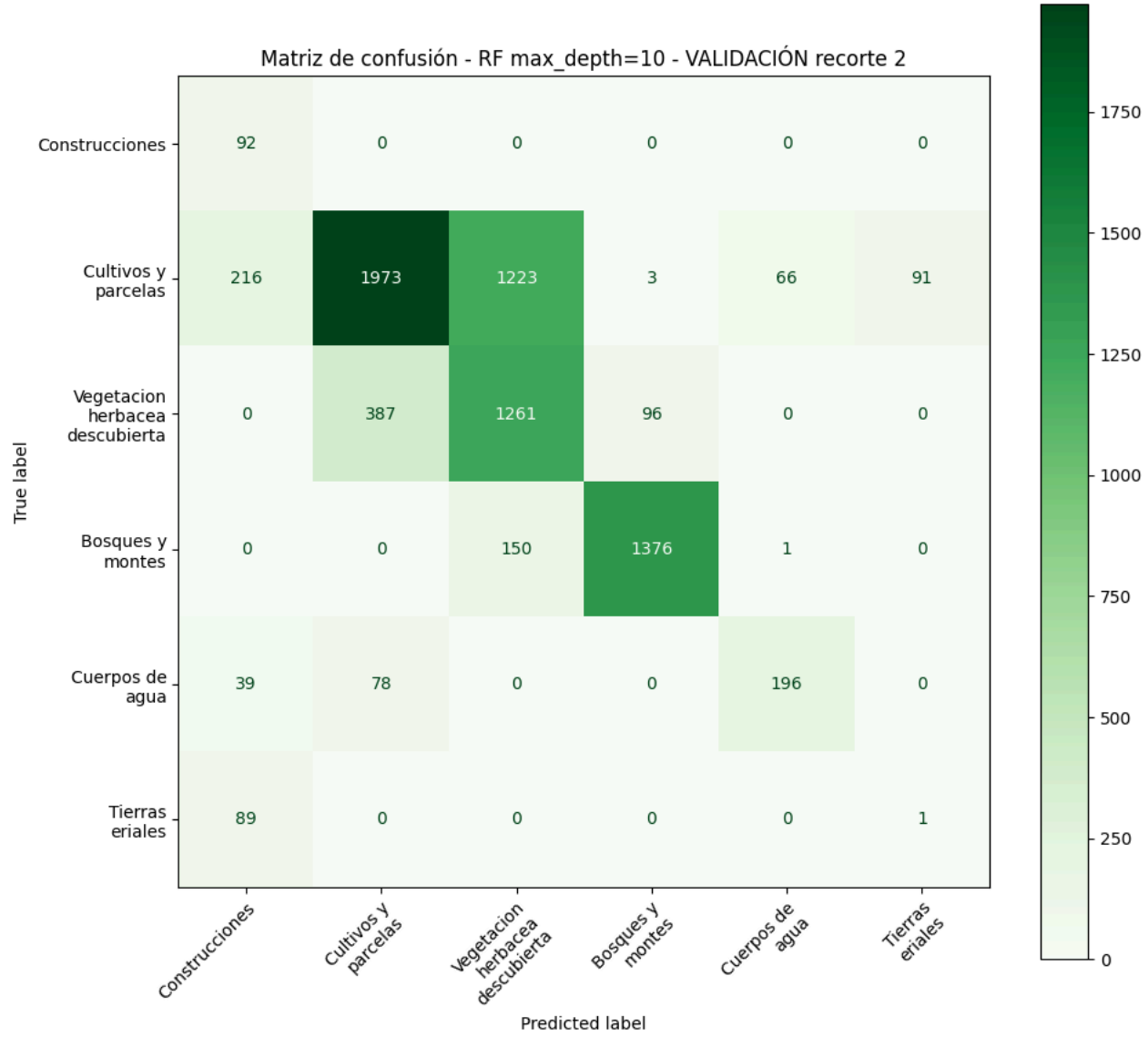



```
----- RF base (max_depth=10) - VALIDACIÓN recorte 2 -----
                              precision    recall  f1-score   support

                Construcciones       0.21      1.00      0.35        92
             Cultivos y parcelas       0.81      0.55      0.66      3572
Vegetacion herbacea descubierta       0.48      0.72      0.58      1744
             Bosques y montes         0.93      0.90      0.92      1527
              Cuerpos de agua         0.75      0.63      0.68       313
              Tierras eriales         0.01      0.01      0.01        90

                    accuracy                           0.67      7338
                   macro avg       0.53      0.64      0.53      7338
                weighted avg       0.74      0.67      0.68      7338
```

```
Precisión global: 0.668
Índice kappa: 0.531
```

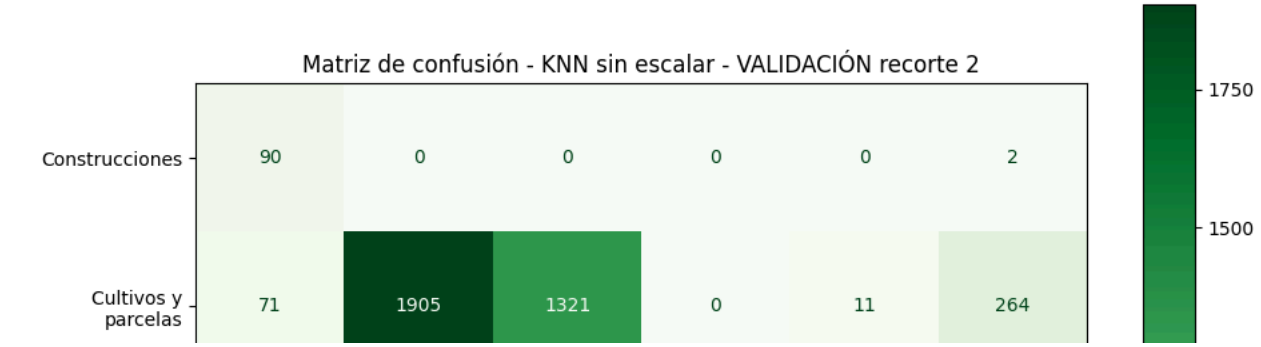


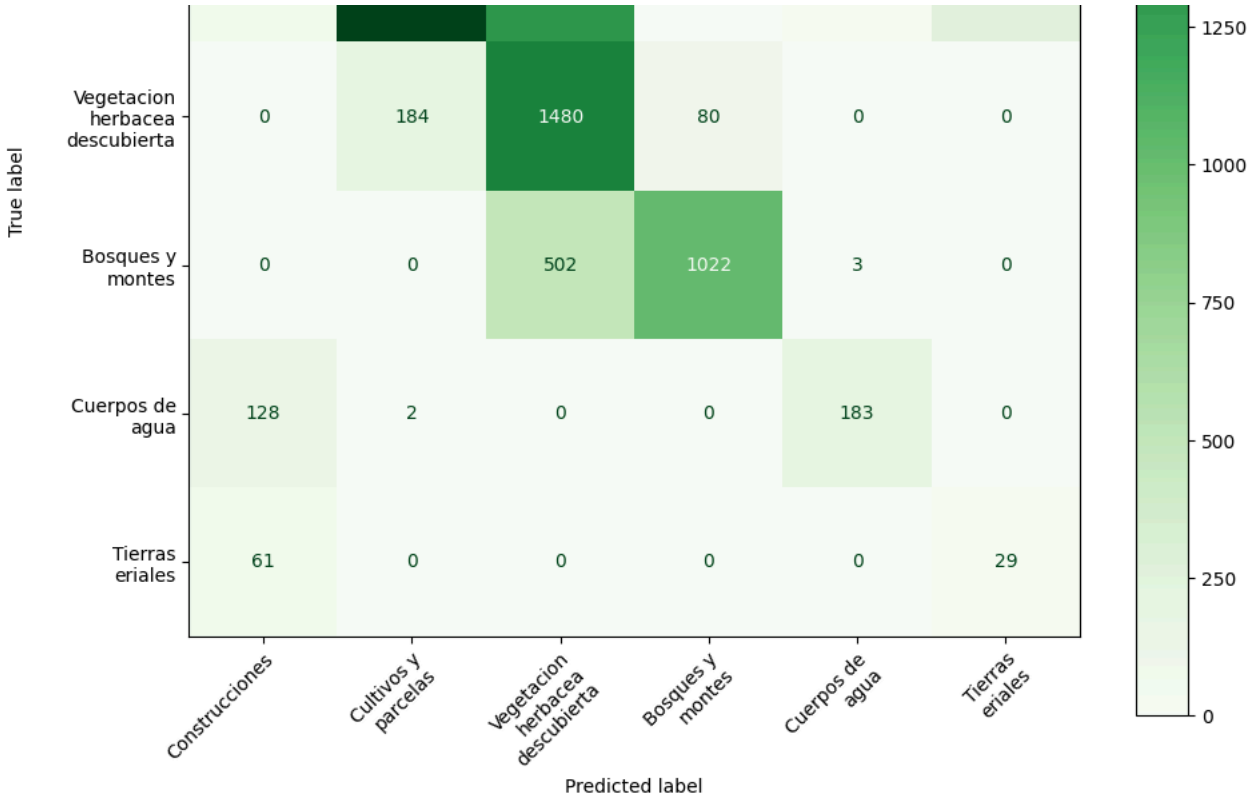

```
----- RF max_depth=10 - VALIDACIÓN recorte 2 -----
                            precision    recall  f1-score   support

                Construcciones       0.21      1.00      0.35        92
            Cultivos y parcelas       0.81      0.55      0.66      3572
Vegetacion herbacea descubierta       0.48      0.72      0.58      1744
              Bosques y montes       0.93      0.90      0.92      1527
             Cuerpos de agua       0.75      0.63      0.68       313
              Tierras eriales       0.01      0.01      0.01        90

                     accuracy                           0.67      7338
                    macro avg       0.53      0.64      0.53      7338
                 weighted avg       0.74      0.67      0.68      7338

Precisión global: 0.668
Índice kappa: 0.531
```



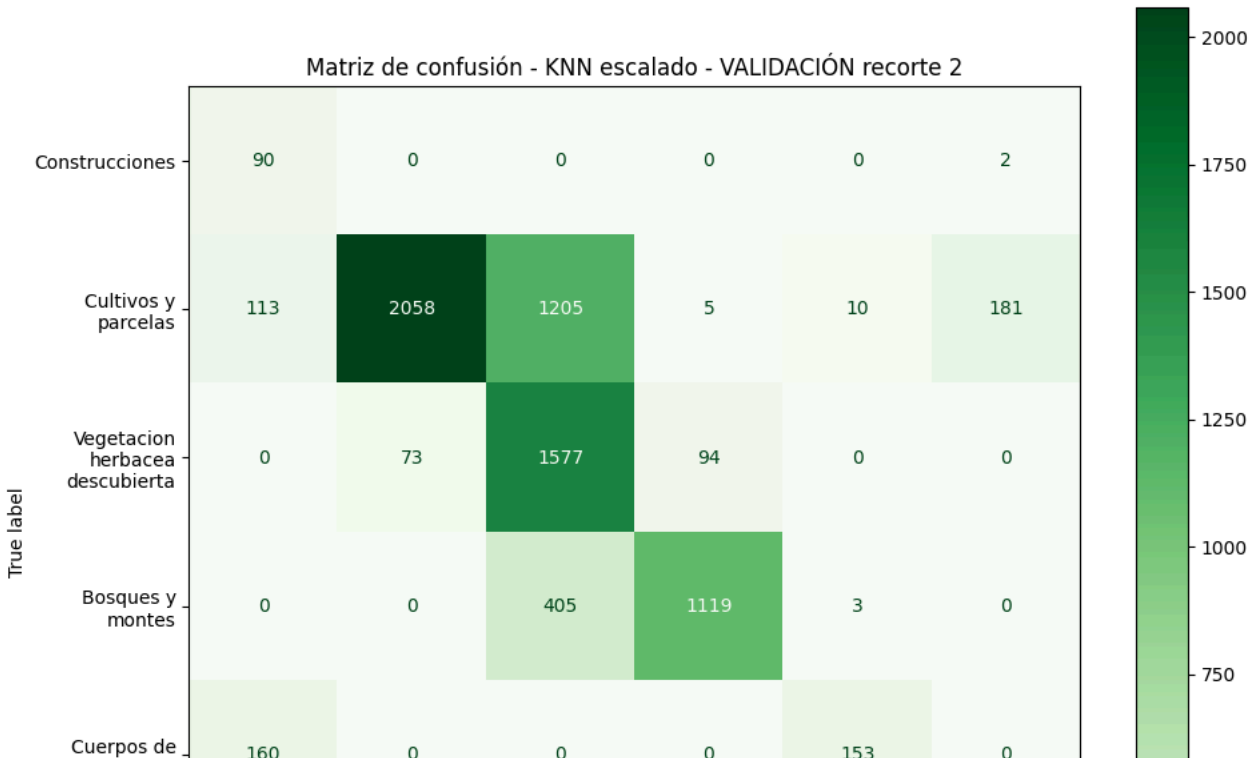


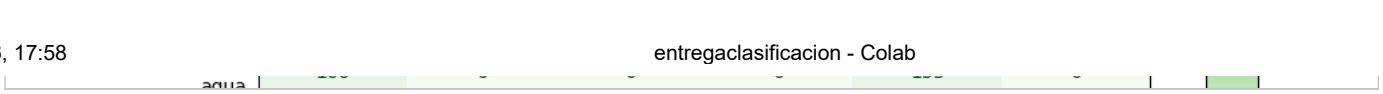

```
----- KNN sin escalar - VALIDACIÓN recorte 2 -----
                             precision    recall  f1-score   support

                Construcciones       0.26      0.98      0.41        92
             Cultivos y parcelas       0.91      0.53      0.67      3572
Vegetacion herbacea descubierta       0.45      0.85      0.59      1744
             Bosques y montes         0.93      0.67      0.78      1527
              Cuerpos de agua         0.93      0.58      0.72       313
              Tierras eriales         0.10      0.32      0.15        90

                     accuracy                           0.64      7338
                    macro avg        0.60      0.66      0.55      7338
                 weighted avg        0.79      0.64      0.67      7338

Precisión global: 0.642
Índice kappa: 0.503
```




agua